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DEPARTMENT OF COMPUTER ENGINEERING

CERTIFICATE

FOR SEMINAR

This is to certify that **Mr. Sahil Bhausheb Nagare** a student of the Department of Computer Engineering, has satisfactorily completed the Seminar work entitled **“Integrating AI and IoT for enhanced Crop monitoring and management”** under the guidance of **Prof. H. V. Thorat** during the Academic Year 2026—27.

Internal Guide

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ABSTRACT

The Internet of Things (IOT) connects a large number of smart sensors, cameras, agricultural equipment, weather stations, and other devices through networks. In modern agriculture, IOT systems can collect important information about soil moisture, temperature, humidity, rainfall, crop conditions, and environmental factors. Artificial Intelligence (AI) can analyze this large amount of agricultural data and provide useful predictions and recommendations for better crop management. This creates an effective approach for improving agricultural productivity and reducing resource wastage. This seminar focuses on integrating Artificial Intelligence (AI) and Internet of Things (IOT) for enhanced crop monitoring and management. The proposed approach involves collecting agricultural data using IOT sensors and cameras, preprocessing the collected data, analyzing crop and environmental conditions using AI and Machine Learning (ML) models, and generating suitable recommendations. AI techniques can be used for crop health detection, disease identification, irrigation prediction, and yield estimation. IOT devices can continuously monitor field conditions and provide real-time data to the AI system. The expected outcome is a smart agricultural monitoring and management system capable of identifying crop-related problems, optimizing irrigation and resources, and supporting farmers in making better decisions.

KEYWORDS

- Internet of Things (IOT)
- Smart Agriculture
- Artificial Intelligence (AI)
- Machine Learning
- Crop Monitoring
- Crop Management
- Soil Moisture
- Crop Disease Detection
- Precision Agriculture
- Sensors
- Computer Vision
- Irrigation Management
- Crop Health
- Yield Prediction
- Data Analysis

INTRODUCTION

The Internet of Things (IOT) is a rapidly growing technology that enables physical devices such as sensors, cameras, weather stations, agricultural machines, and monitoring systems to communicate and exchange data over networks. In agriculture, a large number of IoT devices can be deployed in fields to collect information about soil conditions, weather conditions, crop health, and water requirements. The collected information can help farmers understand the current condition of their crops and make appropriate decisions.

Artificial Intelligence and Machine Learning are increasingly used in smart agriculture to analyze large volumes of agricultural data and identify useful patterns automatically. AI-based systems can analyze sensor readings, images, weather information, and historical crop data to detect crop diseases, predict irrigation requirements, estimate crop yield, and monitor crop health. However, traditional farming methods often depend heavily on manual observation and fixed schedules, which may result in inefficient use of water, fertilizers, and other resources.

Therefore, integrating AI with IoT provides an effective approach for real-time crop monitoring and intelligent agricultural management. IoT sensors continuously collect field data, while AI models analyze the data and generate useful predictions or recommendations. This can help farmers detect problems at an early stage and improve productivity. This seminar examines the integration of AI and IoT for smart agriculture and explores techniques such as sensor-based monitoring, machine learning, computer vision, crop disease detection, irrigation prediction, and data analysis to improve crop monitoring and management.

OBJECTIVES

1. To study and understand the selected problem/domain.

To understand smart agriculture, 10T-based crop monitoring, agricultural sensors, crop-related challenges, and the importance of AI in modern farming.

2. To analyze the relevant technologies and techniques.

To study Artificial Intelligence, Machine Learning, Computer Vision, 10T sensors, soil moisture monitoring, crop disease detection, irrigation prediction, and crop health analysis.

3. To examine the proposed methodology and its expected outcome.

To develop a conceptual methodology in which agricultural data is collected using 10T devices and analyzed using AI models to monitor crop conditions and generate suitable recommendations.

4. To identify practical applications and advantages.

To identify how AI and 10T-based agricultural systems can be applied for smart irrigation, crop disease detection, crop health monitoring, weather monitoring, yield prediction, and efficient resource management.

METHODOLOGY

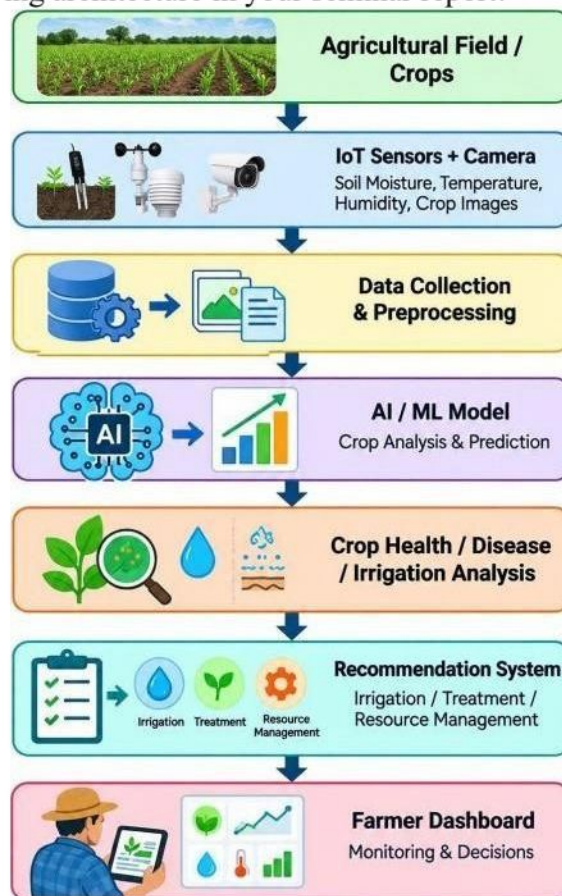
The proposed methodology consists of several stages, beginning with the collection of agricultural data using IoT sensors and cameras. Sensors can collect information such as soil moisture, temperature, humidity, light intensity, and other environmental conditions. Cameras can capture images of crops for monitoring crop health and identifying visible diseases.

The collected data is cleaned and preprocessed to remove irrelevant, noisy, or inconsistent information. Important features are then selected or extracted for analysis by the Artificial Intelligence model. Machine Learning and Computer Vision models can be trained using historical agricultural data and crop images to identify crop conditions, diseases, irrigation requirements, and other agricultural parameters.

After training, the AI model analyzes real-time data received from IoT devices. If the system detects abnormal soil conditions, insufficient moisture, crop disease, or other problems, suitable recommendations can be generated. Based on the analysis, farmers can take appropriate actions such as irrigation, disease treatment, or resource management. Finally, the system can be evaluated using performance metrics such as accuracy, precision, recall, F I score, prediction error, and detection rate.

Methodology / Architecture Diagram

You can draw the following architecture in your seminar report:



Main Techniques

1. IoT Sensors:

IoT sensors collect important agricultural information such as soil moisture, temperature, humidity, light intensity, and environmental conditions from the field.

2. Machine Learning:

Machine Learning models can analyze agricultural data and identify patterns related to crop health, irrigation requirements, disease occurrence, and crop yield.

3. Computer Vision:

Computer Vision can analyze crop images captured using cameras or mobile devices and identify visible symptoms of crop diseases and unhealthy plants.

4. Crop Disease Detection:

AI models can be trained using crop images to identify different diseases and classify healthy and unhealthy crops.

5. Smart Irrigation:

Soil moisture sensors and AI models can be used to determine when irrigation is required and help reduce unnecessary water usage.

6. Crop Prediction:

AI models can analyze historical and real-time agricultural data to predict crop growth, yield, and possible agricultural conditions.

ADVANTAGES

Real-time crop monitoring: IOT sensors can continuously collect information about crop and soil conditions.

Early disease detection: AI-based image analysis can help identify crop diseases at an early stage.

Efficient water usage: Smart irrigation systems can provide water according to actual soil moisture requirements.

Improved crop productivity: Continuous monitoring and intelligent recommendations can support better crop management.

Reduced manual effort: Automated monitoring reduces the need for continuous manual observation of fields.

APPLICATIONS

1. Smart Irrigation

IOT soil moisture sensors can monitor the amount of moisture present in the soil. AI can analyze the sensor data and determine whether irrigation is required.

2. Crop Disease Detection

Cameras can capture crop images and AI-based computer vision models can analyze these images to identify diseases and unhealthy plants.

3. Crop Health Monitoring

IOT sensors and AI models can continuously monitor environmental and crop conditions. Farmers can receive information about crop health through a dashboard.

4. Weather Monitoring

Temperature, humidity, rainfall, and other environmental conditions can be collected using IOT devices. This information can be analyzed to support agricultural decisionmaking.

5. Yield Prediction

AI models can analyze historical crop data, weather conditions, soil information, and other parameters to estimate expected crop yield.

6. Fertilizer Management

Agricultural data can be analyzed to identify crop and soil requirements and support efficient fertilizer application.

EXPECTED OUTCOME

The proposed approach is expected to produce an intelligent AI and IoT-based agricultural monitoring and management system that can continuously monitor crop and environmental conditions while providing useful recommendations to farmers. The system is also expected to demonstrate the importance of combining real-time IoT data with Artificial Intelligence for improving agricultural decision-making.

The performance of the proposed model can be evaluated using:

Metrics	Purposes
Accuracy	Measures overall correct predictions
Precision	Measures how many predicted diseases are actually diseases
Recall	Measures how many actual diseases are detected
F1-Score	Provides a balance between precision and recall
False Positive Rate	Measures the difference between predicted and actual values
Detection Rate	Measures the ability to identify crop problems

CONCLUSION

The increasing use of IoT devices in agriculture has created new opportunities for real-time crop monitoring and efficient agricultural management. Traditional farming methods often depend on manual observation and fixed schedules, which can result in inefficient use of water, fertilizers, and other resources. Artificial Intelligence and Machine Learning can analyze agricultural data collected through IoT devices and provide useful predictions and recommendations.

AI models can be used for crop health monitoring, disease detection, irrigation prediction, yield estimation, and resource management. IoT sensors can continuously collect information about soil and environmental conditions, while AI models can analyze this information and support better agricultural decisions. The proposed approach focuses on developing an intelligent crop monitoring and management system that can improve productivity, reduce resource wastage, and support farmers.

Such systems have applications in smart irrigation, crop disease detection, weather monitoring, precision agriculture, fertilizer management, and yield prediction. In the future, this work can be extended using real-time edge computing, drones, robotics, satellite imagery, advanced AI models, and automated agricultural systems to improve the efficiency, accuracy, and scalability of smart farming.

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